GCP Migration Program Approach

20 applications, two platforms, 26 weeks

Scenario 1 · Program planning

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All 20 applications live by week 22

Replatform, do not redesign. Applications move in three waves ordered by dependency and risk, each using one repeatable cutover pattern. Resilience is tested inside every wave, not left to the end.

What I commit to as program owner

- 20 applications and their databases in production on GCP by week 22
- Recovery targets agreed in week 3 and demonstrated by week 24
- A gate at the end of every milestone; a go/no-go and a tested rollback for every cutover
- Any threat to the end date raised within a week of discovery

20

applications across two platforms

3

waves: pilot, Platform A, Platform B

Week 24

HA and DR proven in a full exercise

2 weeks

contingency held at program level

Holds on three conditions: no material application refactoring · key decisions closed by week 4 · committed team capacity

What the brief leaves open, and what I assumed

Open question	Assumption	If wrong	Confirmed by
Why are the platforms complex?	Shared databases and batch jobs; integrations cross both platforms	Larger move groups; fewer, bigger cutovers	Traffic analysis, week 3
Can applications move as they are?	On-premises source; 16 of 20 containerize without code change; the rest rehost on VMs	Each refactor adds about 3 weeks	Fitness review, week 3
How critical, what recovery?	6 Tier 1 (RTO 1 hour, RPO 5 minutes), 9 Tier 2 (4 hours, 15 minutes), 5 Tier 3 (24 hours)	Tighter targets need active-active	Business sign-off, week 3
What regulation applies?	Regulated financial data; US regions only; audit evidence at every gate	A new approval step before G2	Security and risk review, week 2
How much downtime?	Tier 1: one 4-hour weekend window. Tier 2 and 3: a weeknight window	Zero downtime needs another pattern	Business sign-off, week 4
Who does the work?	Two squads of six, a platform team of five, 30 percent of each application team	A third squad or less scope is needed	Sponsor commitment, week 4
What constrains the calendar?	One change freeze of up to two weeks, assumed in weeks 12 to 13 when no cutovers are planned; dual running is funded	A freeze elsewhere uses contingency	Change calendar, week 4

Replatform onto managed GCP services

Layer	Target
Java and modern .NET	Containers on regional GKE clusters
Legacy .NET Framework	Windows VMs in regional managed instance groups
PostgreSQL	Cloud SQL, regional HA, cross-region replica
Data migration	Database Migration Service, continuous replication
Network	Shared VPC, interconnect, VPN as backup
Delivery and operations	One pipeline, infrastructure as code, central monitoring

Disaster recovery by tier

Tier 1 · 6 applications

Warm standby in a second region

RTO 1 hour · RPO 5 minutes

Tier 2 · 9 applications

Pilot light: live replica, compute rebuilt from code

RTO 4 hours · RPO 15 minutes

Tier 3 · 5 applications

Restore from cross-region backup

RTO 24 hours · RPO 24 hours

All tiers run across multiple zones with automatic database failover. The 5-minute RPO is measured under load in the pilot, because the cross-region replica is asynchronous.

One cutover pattern, repeated 20 times

1

Build

T minus 3 weeks

Infrastructure from code; application in non-production; replication started

2

Test

T minus 2 weeks

Functional, integration, performance and security tests on the target

3

Rehearse

T minus 1 week

Full cutover and rollback with production-sized data, timed

4

Cut over

T

Go/no-go, stop writes, final sync, reconcile data, switch traffic, validate

5

Hypercare

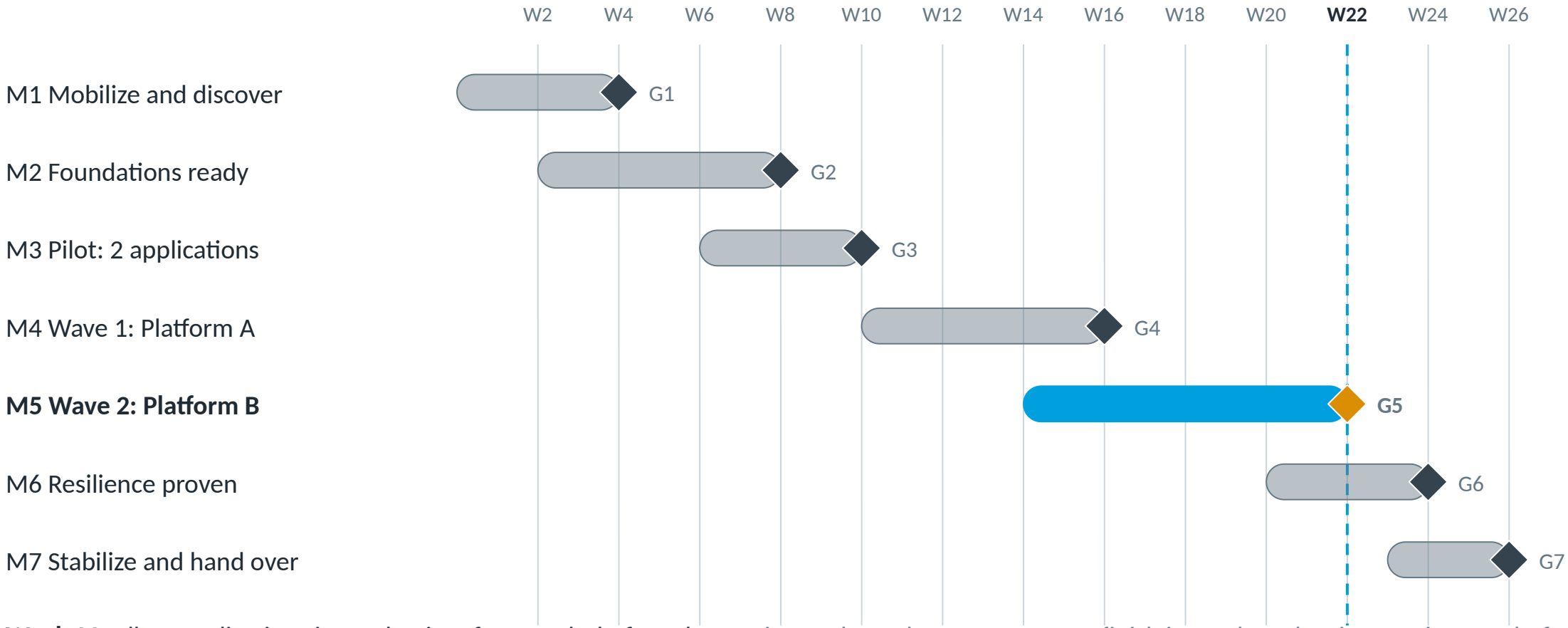
T plus 2 weeks

Squad stays on call; source kept read-only for 30 days for reconciliation, not failback

Rollback: a decision point at T plus 2 hours. Before it, traffic returns to the source. After it, the team fixes forward.

A common pattern is what makes 20 cutovers in 13 weeks credible: every team learns it once, and each rehearsal produces a measured duration before any go decision.

Seven milestones over 26 weeks



Week 22: all 20 applications in production, four weeks before close. Diamonds mark gates. Cutovers finish in week 21, leaving contingency before G5 and before G7.

Critical path: interconnect and security sign-off, G2, pilot cutover, G3, Wave 1, Wave 2 Tier 1 cutovers, DR exercise, G6.

M1 Mobilize and discover: weeks 1 to 4

Replace assumptions with facts and baseline a plan the sponsors sign.

Activity	Owner	Weeks	Output
Confirm sponsor, governance and decision rights	Program lead	1	Charter, responsibility matrix
Order interconnect; open security baseline review	Platform and security leads	1	Order placed; review booked
Inventory 20 applications and their dependencies	Application leads	1 to 3	Inventory, dependency map
Decide containerize or rehost; check databases	Platform and database leads	2 to 3	Disposition per application
Agree tiers and recovery targets	Program lead, business owners	3	Signed tiering
Define move groups, wave plan, capacity, budget	Program lead	3 to 4	Baselined plan
Approve target architecture and DR design	Cloud platform lead	4	Architecture decision record

Exit gate: G1 Baseline approved, week 4: inventory and dependency map complete; tiers and recovery targets signed; plan and budget baselined.
Lever if late: baseline Wave 1 only and finish Platform B discovery during foundations.

M2 Foundations ready: weeks 3 to 8

A secure, observable landing zone that any application can be deployed to from code.

Activity	Owner	Weeks	Output
Build landing zone as code in two regions	Cloud platform lead	3 to 6	Landing zone
Bring up VPN, then interconnect	Cloud platform lead	4 to 7	Hybrid connectivity
Build CI/CD pipeline and base images	Cloud platform lead	4 to 7	One pipeline, three templates
Define database, replication and backup patterns	Database lead	5 to 7	Reusable database module
Set up logging, monitoring and alerting	SRE lead	5 to 8	Operations baseline
Security review and penetration test	Security lead	7 to 8	Sign-off
Deploy a reference application end to end	Cloud platform lead	8	Working proof

Exit gate: G2 Foundations accepted, week 8: security sign-off; interconnect live; pipeline deploys a reference application.

Lever if late: the pilot proceeds over VPN; Wave 1 does not start without the interconnect.

M3 Pilot, 2 applications: weeks 7 to 10

Prove the cutover pattern on one .NET and one Java Tier 3 application, and measure what was only estimated.

Activity	Owner	Weeks	Output
Build both applications in non-production	Squad 1	7	Target environments
Functional, integration and performance tests	Test lead	8	Test report
Rehearse cutover and rollback, timed	Squad 1, database lead	8 to 9	Measured durations
Production cutover, after G2	Program lead runs go/no-go	9	Two applications live
Zone failover test; DR dry run; replica lag	SRE lead	9 to 10	Resilience evidence
Retrospective; update runbook and estimates	Program lead	10	Runbook version 1; re-forecast

Exit gate: G3 Pilot live, week 10: two applications in production; cutover and rollback rehearsed; zone failover passed; re-forecast issued.
Lever if late: effort more than 25 percent over estimate goes to steering as a re-plan.

M4 Wave 1, Platform A: weeks 11 to 16

Move the remaining 8 Platform A applications, reaching 10 of 20 in production.

Activity	Owner	Weeks	Output
Build 8 applications; start replication	Squads 1 and 2, Platform A lead	11 to 12	Target environments
Test each move group	Test lead	12 to 14	Test reports
Rehearse each cutover and rollback	Squads, database lead	13 to 15	Timings inside the window
Cut over Tier 3 and Tier 2 on weeknights	Program lead runs go/no-go	14 to 15	5 applications live
Cut over Tier 1 at weekends	Program lead runs go/no-go	15 to 16	3 applications live
Zone failover test per application	SRE lead	14 to 16	HA evidence

Exit gate: G4 Platform A complete, week 16: 10 applications in production; no open Severity 1 or 2 defects.
Lever if late: a move group that fails rehearsal joins the Wave 2 schedule and does not delay the wave.

M5 Wave 2, Platform B: weeks 15 to 22

Move all 10 Platform B applications, reaching 20 of 20 in production.

Activity	Owner	Weeks	Output
Build 10 applications; start replication	Squads 1 and 2, Platform B lead	15 to 18	Target environments
Test each move group	Test lead	16 to 19	Test reports
Rehearse each cutover and rollback	Squads, database lead	17 to 20	Timings inside the window
Cut over Tier 3 and Tier 2 on weeknights	Program lead runs go/no-go	18 to 20	7 applications live
Cut over Tier 1 at weekends	Program lead runs go/no-go	20 to 21	3 applications live
Source set read-only and retained; contingency before G5	Program lead	22	Evidence for gate G5

Exit gate: G5 All applications live, week 22: 20 applications in production; source read-only and retained.

Lever if late: the contingency before G5 absorbs one slip. A second slip goes to steering: add a squad, or defer a Tier 3 application.

M6 Resilience proven: weeks 21 to 24

Demonstrate, with evidence, that the target meets its recovery targets.

Activity	Owner	Weeks	Output
Approve DR exercise scenario and criteria	SRE lead, business owners	21	Exercise plan
Load test the full estate at peak volume	Test lead	21 to 22	Performance report
Regional failover exercise, Tier 1 and 2	SRE lead	23	Measured RTO and RPO
Timed backup restore for Tier 3	Database lead	23	Measured restore time
Fail back; fix findings; retest	SRE lead	24	Closed findings
Assemble evidence pack for risk and audit	Program lead	24	Evidence pack

Exit gate: G6 Resilience proven, week 24: DR exercise meets recovery targets; performance within 10 percent of baseline.
Lever if late: a missed target is fixed and retested for that application only.

M7 Stabilize and hand over: weeks 24 to 26

Operations owns the platform, and the program can close.

Activity	Owner	Weeks	Output
Hypercare for the final wave; defect burn-down	Squads 1 and 2	24 to 25	No open Severity 1 or 2
Hand over runbooks, on-call and dashboards	SRE lead	24 to 25	Signed handover
Set cost baseline and optimization backlog	Finance partner, platform lead	25	Run-cost baseline
Agree dated decommission plan for the source	Program lead	25	Approved plan
Closure report and lessons learned	Program lead	25 to 26	Closure report

Exit gate: G7 Program closed, week 26: operations accepts handover; decommission plan approved.
Lever if late: low-severity defects move to application team backlogs with named owners.

The work fits, with 18 percent headroom

140

person-weeks of demand

20 applications at 7 person-weeks each

171

person-weeks of supply

228 gross, 75 percent productive

31

person-weeks of headroom

about four applications of slack

Supply	Calculation	Person-weeks
Squad 1	6 people, weeks 5 to 25	126
Squad 2	6 people, weeks 9 to 25	102
Gross		228
Productive	75 percent, after onboarding, hypercare and DR exercise support	171

The 7 person-week estimate is the assumption the pilot tests. At 25 percent over, demand reaches 175 and exceeds supply. That is why a 25 percent overrun at gate G3 goes to the steering committee as a re-plan.

The platform team of five and the 30 percent contribution from application teams are in addition, and are not counted here. Peak load is three to four cutovers a week in weeks 14 to 15 and 19 to 20.

Six measures, each with a target, trigger and response

Measure	Target	Trigger	Response	Decided by
Applications live against plan	On or ahead of plan	Two or more behind	Re-sequence move groups; then draw contingency	Program lead, then steering
First-time cutover success	At least 90 percent	Any rollback	Cause fixed before the next similar cutover	Program lead
Rehearsal inside the window	All, before any go	Any rehearsal over its window	No go. Re-tune or request a longer window	Database lead, business owner
Severity 1 and 2 in hypercare	Zero at each gate	Any open at a gate	The gate does not pass	Steering committee
Recovery targets in test	All Tier 1 and 2 by week 24	Any target missed	Fix and retest that application only	SRE lead
Spend against plan	Within 5 percent	More than 5 percent over	Options to steering within a week	Program lead, finance partner

Rhythm: daily wave stand-up, weekly program review of these measures, go/no-go before every cutover, steering every two weeks, stakeholder report monthly.

Six risks most likely to threaten the date

Risk	Likelihood / impact	What this plan does about it	Owner	Seen first-hand
Undocumented dependencies surface at cutover	**High / High**	Traffic analysis in discovery, covering vendor and external calls; not interviews alone	Application leads	**An external vendor API that nobody had listed**
Application teams pulled back to other work	**High / High**	Capacity is a sponsor decision by week 4; regulatory and freeze dates are on the wave plan	Executive sponsor	**A regulatory deadline took the team**
Database cutover exceeds its window	**Medium / High**	Every cutover rehearsed with production-sized data and timed; over the window means no go	Database lead	**A migration overran by over a week**
Security or audit sign-off arrives late	**Medium / High**	Security lead embedded from week 1; patterns approved once; evidence at every gate	Security lead	**Sign-off arrived late**
Legacy .NET cannot be containerized	**Medium / Medium**	Decide per application by week 3; rehost on virtual machines	Cloud platform lead	**A legacy app needed rework before moving**
DR exercise fails late in the program	**Low / High**	Zone failover in every wave; DR dry run in the pilot	SRE lead	**A DR test missed its recovery target**

Five decisions and what would change them

Decision	Options considered	Choice and why	Would change if
Migration strategy	Rehost all; replatform; refactor	Replatform. Managed services give HA and DR without code change	More than four applications cannot run unchanged
Sequencing	By tier; by platform; one cutover	A pilot, then one platform at a time. Least hybrid latency and coordination	Discovery shows heavy coupling across the platforms
Database target	Cloud SQL; AlloyDB; PostgreSQL on VMs	Cloud SQL. Built-in failover and the smallest operational change	An extension or performance gap appears in week 3
DR posture	Active-active; warm standby; backup only	Tiered: warm standby, pilot light, restore. Meets a 1-hour RTO without redesign	The business needs near-zero RTO for any application
Cutover style	Zero downtime; planned window	A planned window with a timed rehearsal. Simple, with a clean rollback	A Tier 1 application has no acceptable window

Each choice trades modernization for speed. Refactoring and active-active are logged as a backlog for after the program, not hidden.

Decisions needed, and what gives first

Decision	Owner	Week
Recovery targets per tier	Business owners	3
Containerize or rehost, per application	Cloud platform lead	3
Primary and DR regions	Platform and security leads	3
Team capacity and roadmap reductions	Executive sponsor	4
Cutover windows and freeze calendar	Business owners	4
Rollback point of no return per tier	Program and database leads	9
Source decommission dates	Executive sponsor	25

The date is fixed. In order, I would:

1. Re-sequence move groups inside a wave (my call)
2. Draw on the two contingency weeks (steering)
3. Add a third squad for Wave 2 (sponsor)
4. Defer Tier 3 DR automation (steering)
5. Defer a Tier 3 application (sponsor)

Week 10 is the first point where the date is confirmed with evidence. I would say so at kick-off.